Fatmawati Traffic Dataset Explained

Oct 8, 2026 · @Daffa

Overview

`data/fatmawati_v3` is a synthetic, sensor-only dataset: 108 recorded hours of what roadside loops, counting gates and queue cameras would log at Perempatan Fatmawati (Cilandak, South Jakarta), produced by a SUMO 1.27.1 digital twin of the junction and the main roads within 2.5 km. Generator `traffic_twin_fatmawati_v3.0`, plan `release`, generated on 2026-10-04 and committed on 2026-10-06.

It exists because the junction sits 326 m north of RSUP Fatmawati, a national referral hospital with a 24-hour emergency department, and four in five vehicles are motorcycles that fill the motorcycle box and every gap at red. An ambulance gets through only when its approach is given a timely green, and such passages cannot be recorded safely on live roads, so the twin generates the sensor data a priority rule could be learned from.

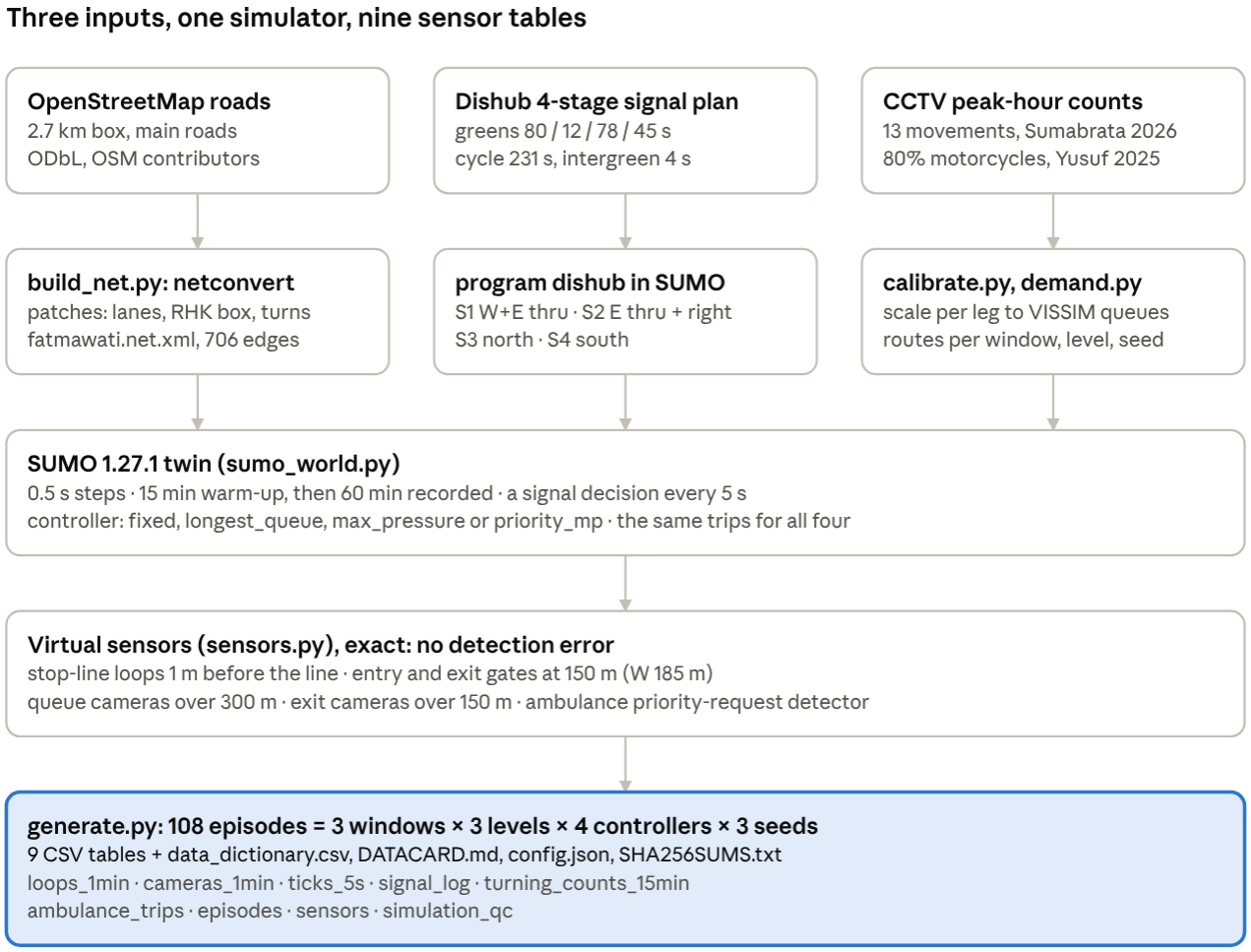
It answers three questions: how four signal controllers (Dishub fixed plan, longest queue, max-pressure, and the ambulance-aware teacher `priority_mp`) compare under identical demand; what the teacher would request at each 5 s decision instant (label `y_teacher_stage`); and how long each approach queue will be 30 s ahead (labels `y_q_veh_30s_N/E/S/W`). Every episode also carries ambulance trips to RSUP Fatmawati, which is the healthcare framing of the ITI5145 project.

The release is nine CSV tables (423,907 rows, 42.7 MB) plus `data_dictionary.csv`, `DATACARD.md`, `config.json` and `SHA256SUMS.txt`. No vehicle trajectories, driver parameters or simulator states are written out; vehicle identities were used only inside the simulator to match gate entries with exits. The road network derives from OpenStreetMap, so the dataset is shared under the Open Database License (ODbL 1.0).

The older `data/traffic_v1` release (a 2x2 synthetic grid, generator v1.2) is still in the repository and feeds the v1 ML pipeline, but it is a different dataset and is not described here. The other folders under `data/` are inputs or by-products: `fatmawati` holds the network, patches and calibration, `fatmawati_pilot` a two-episode trial with the same file names, and the five `fatmawati_eval*` folders the ambulance evaluation grids. The release is tracked in the project's private GitHub repository and has no public download link yet.

Provenance: how the data was generated

Nothing in the tables was measured on the road. Every row is what a virtual sensor recorded inside a SUMO simulation whose geometry, signal plan and demand pattern come from three real sources.



generation pipeline · from docs/DECISIONS.md §13 and traffic_twin/fatmawati/

The network, the signal program and the demand are built once, then one simulation runs per episode; the virtual sensors watch it and the generator writes their readings out.

Demand. Each of the 13 movements counted at the junction enters at the upstream end of its leg and leaves at the far end of its exit leg, routed through the entry and exit gates so the movement happens at Fatmawati. Per approach the counted volume is multiplied by a calibration scale (N 0.242, E 0.239, S 0.220, W 0.360), by the level factor (low 0.75, design 1.0, high 1.15) and by the hour's share of the daily-profile peak for the window (07:00, 12:00 or 17:00). Background trips between the fringes stay 200 m from the junction. The random seed fixes the trips, so within one window, level and seed the four controllers face identical traffic and controller comparisons are paired.

Recording. The simulation runs 15 minutes of warm-up so the network fills, then records 60 minutes. Simulated time `t` keeps counting from the start, so recorded rows begin at `t = 900`; `clock` is derived from the window's start hour. Warm-up rows are dropped from every table, and the simulation runs 30 s past the hour so that every recorded tick has its 30 s queue label. Ambulance trips that are still travelling at the recorded hour's end are followed for up to 15 more minutes so their arrival can be logged.

Files and schema

Nine CSV tables share the key `episode_id`; the time-series tables add `t` or `t_start` (simulated seconds since the episode began, warm-up included, so recording starts at 900) and a derived `clock` string. Sensor tables reference `sensors.csv` through `sensor_id`. Every column's unit, role and meaning is in `data_dictionary.csv` (206 rows).

File (rows)	One row per; keys	What it holds
`sensors.csv` (39)	sensor; `sensor_id`	kind (stop-line loop, entry gate, exit gate, queue camera, exit camera), leg, lane, street, WGS84 lat/lon, distance from the junction centre, camera zone length
`episodes.csv` (108)	episode; `episode_id`	scenario (window, level, seed, controller, warm-up and recorded duration) and sensor-derived outcomes: stop-line counts, mean queue and stopped delay per approach, junction stopped delay, fairness ratio, entry volume, motorcycle share, cordon travel time, ambulance trips, travel time and recovery

File (rows)	One row per; keys	What it holds
`loops_1min.csv` (200,880)	loop x minute; `episode_id`, `t_start`, `sensor_id`	counts by class (motorcycle, car, bus, truck, emergency), total, PCU, occupancy %, mean speed at the loop
`cameras_1min.csv` (51,840)	camera x minute; `episode_id`, `t_start`, `sensor_id`	mean and max standing vehicles, standing PCU, mean and max queue length (m), vehicles in zone, space-mean speed, motorcycles in the box, seconds with an ambulance present
`ticks_5s.csv` (77,760)	junction x 5 s decision instant; `episode_id`, `t`	the ML table: 85 columns of per-approach features, stage pressures, signal state, the teacher label and four 30 s queue forecast labels (families below)
`signal_log.csv` (77,202)	phase change; `episode_id`, `t`	phase index, stage or stage change (e.g. `S4>S1`), state (green, amber, all-red), controller
`turning_counts_15min.csv` (15,238)	15 min x entry leg x exit leg x vehicle class; `episode_id`, `t_start`, `entry_leg`, `exit_leg`, `vclass`	vehicles exited, mean, median and 85th-percentile entry-to-exit travel time (re-identified at the gates)
`ambulance_trips.csv` (732)	ambulance trip; `episode_id`, `trip`	AVL departure and arrival, first detection leg, distance and queue at detection, gate, stop-line and exit times, recovery time, travel time, cordon time, arrived flag

File (rows)	One row per; keys	What it holds
`simulation_qc.csv` (108)	episode; `episode_id`	engine diagnostics, not sensor readings: lane collisions, junction overlaps, teleports, vehicle-km, departed, arrived, unmatched exits, wall time, SUMO version
`data_dictionary.csv` (206)	column; `table`, `column`	unit, role (key, metadata, scenario, outcome, reading, feature, label, qc) and description
`config.json`	the release	generator version, plan, windows, level factors, calibration scales, vehicle mix, PCU factors, gate radii, camera lengths, label horizon, network SHA-256
`DATACARD.md`	the release	the data card: calibration, sensors, episodes, row counts, simulation quality, licence
`SHA256SUMS.txt` (12)	file	checksums of every file above

Column families in `ticks_5s.csv`

The same 15 readings repeat for each approach leg N, E, S and W, so one row is one snapshot of the whole junction.

Family	Columns per leg	Meaning
Queue camera (QC)	`q_veh`, `q_pcu`, `q_m`, `moto_box`, `cam_speed_kmh`	standing vehicles now (< 0.1 m/s), the same in PCU, queue length from the stop line to the last standing vehicle, motorcycles within 8 m of the stop line, space-mean speed (empty when the zone is empty)

Family	Columns per leg	Meaning
Loops and gates	`dis_5s`, `dis_60s`, `arr_60s`	vehicles over the stop-line loops in the last 5 s and 60 s, vehicles through the entry gate in the last 60 s
Exit camera	`exit_q_pcu`	standing PCU on the exit leg (spill-back)
Priority request	`ev`, `ev_dist_m`, `ev_eta_s`, `ev_q_ahead_pcu`, `ev_needs_signal`	ambulance seen in the approach zone, its distance, ETA at no less than 5 m/s, standing PCU ahead of it, whether its route crosses the stop line
Pressure	`pressure`	max-pressure of the approach: standing PCU minus exit queues weighted by the turning shares counted so far
Lane groups (E only)	`q_veh_E_S1S2`, `q_veh_E_S2`	standing vehicles on the E lanes that go through (green in S1 and S2) and that turn right (S2 only)
Stages	`stage_q_S1..S4`, `stage_pressure_S1..S4`	standing vehicles and pressure each of the four stages would serve
Signal state	`stage`, `phase_state`, `transition_to`, `t_in_phase_s`, `t_since_green_s`	the running stage, green/amber/all-red, the stage being changed to, timers
Metadata	`controller`, `controller_request`	the running controller and the stage it requested at its last decision; `controller_request` is, under `priority_mp`, the teacher's own request held since the last decision instant, which matches the label on about two-thirds of ticks, so it must be excluded from features

Family	Columns per leg	Meaning
Labels	`y_teacher_stage`, `y_q_veh_30s_N/E/S/W`	the stage the priority-aware teacher requests now; standing vehicles per approach 30 s later (the simulation runs 30 s past the recorded hour, so every tick has one)

Passenger-car units follow PKJI 2023: motorcycle 0.15, car 1.0, bus and truck 1.3, emergency 1.0. Legs map to streets as N and S = Jl. RS Fatmawati, E = Jl. TB Simatupang, W = Jl. RA Kartini.

Coverage

The release is a full factorial of 3 windows x 3 demand levels x 4 controllers x 3 seeds, one episode per cell, so every comparison across controllers is paired and every comparison across levels or windows has three seeds behind it.

Dimension	Values	What it changes
Window	am_peak 07:00, midday 12:00, pm_peak 17:00	only a volume multiplier from the daily profile: 0.86, 0.64 and 0.93 of the calibrated peak hour; turning shares, vehicle mix and the ambulance rate are identical in every window
Level	low, design, high	0.75, 1.0 and 1.15 x the calibrated volumes; realised mean entry volume 4,165, 5,472 and 6,061 veh/h
Controller	fixed, longest_queue, max_pressure, priority_mp	the signal logic only; within a window, level and seed all four face the same trip file
Seed	1, 2, 3	the random trips and the simulation; the calibration check used seeds 11 to 14, so no release seed was used for fitting

Sensors and corridor. 39 sensors sit on the four legs within 185 m of the junction centre at 6.2925 S, 106.7952 E: 11 stop-line loops (N 3 lanes, E 4, S 2, W 2), 11 entry-gate and 9 exit-gate loops at 150 m (185 m on Jl. RA Kartini), four queue cameras covering 300 m of each

approach and four exit cameras covering 150 m of each exit. Only the junction's own approaches are instrumented; the rest of the 2.7 km network exists to feed and drain them.

Rows per recorded hour. Each episode contributes 720 ticks, 1,860 loop-minutes (31 loops x 60 min), 480 camera-minutes, 125 to 155 turning-count rows (13 movements x vehicle classes x four 15 min periods) and 1 to 11 ambulance trips. The signal log has exactly 189 rows under the fixed plan (63 greens of a 231 s cycle) and 669 to 1,064 under the adaptive controllers, whose greens last 5 to 90 s with a median of 5 s.

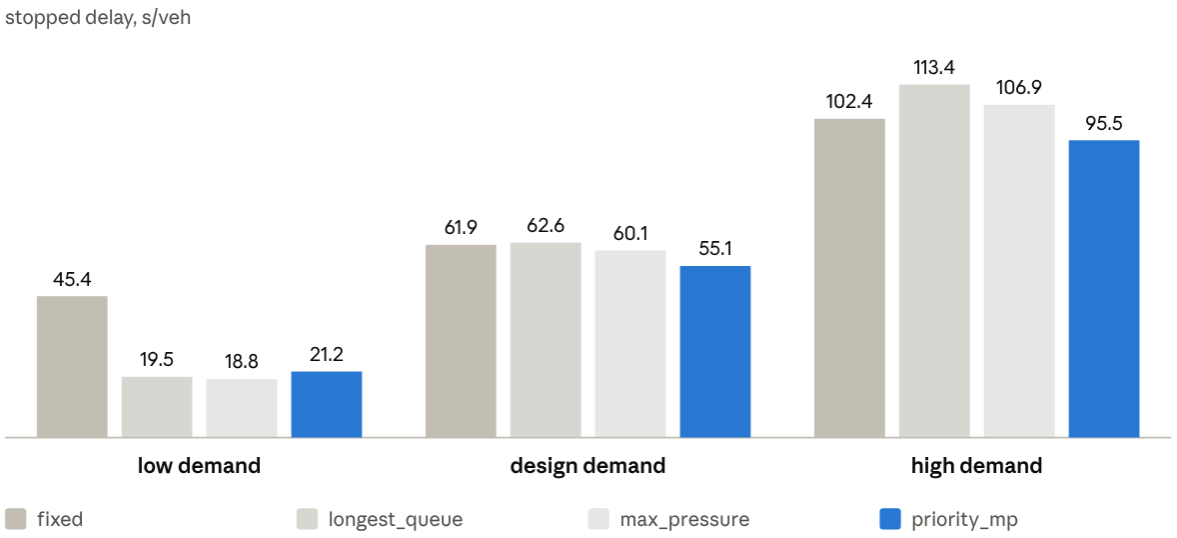
Traffic. About 7,800 vehicles are inserted per hour across the network; 544,861 gate passages were matched entry to exit over the release, 80.1% of them motorcycles. All 13 counted movements appear, including the U-turns E to E and W to W (16% of exits); the removed Jl. RA Kartini right turn (W to S) never does. Standing vehicles across the four approaches average 20 at low, 60 at design and 118 at high demand, with instants of up to 580; the north queue reaches the 300 m camera edge in 2.8% of ticks and in 11% of ticks at high demand.

Labels and ambulances. `y_teacher_stage` is S3 (north) on 37% of ticks, S1 on 30%, S4 on 21% and S2 on 11%, and differs from the running stage on 69% of ticks. An ambulance is inside an approach camera zone on 13.6% of ticks (N 6.2%, E 5.9%, W 2.7%, never S). Of the 732 trips, 724 reached the hospital within the simulated time; median travel time 358 s, 85th percentile 686 s; 406 trips crossed a signalised stop line (N 200, W 206) and 326 used the unsignalised east free-left slip.

Missing values are structural only. Loop speed is empty when no vehicle crossed in the minute (14.5% of loop-minutes); the ambulance request fields are empty when no ambulance is on the approach; `moto_in_box_mean` is empty for exit cameras; `t_stopline` and `t_recovered` are empty for slip-road trips; `ambulance_recovery_mean_s` is empty in 8 episodes. No other column has gaps.

Adaptive controllers halve stopped delay at low demand, not at high demand

Mean junction stopped delay per episode in s/veh, 9 episodes per bar (3 windows × 3 seeds)

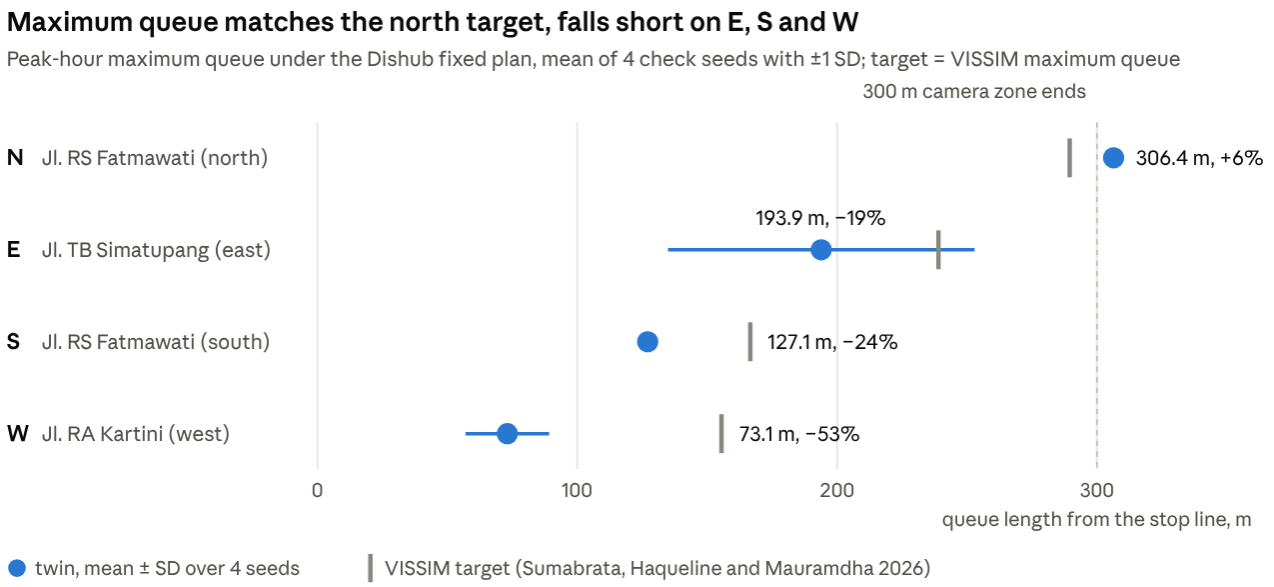


data/fatmawati_v3/episodes.csv · stopped_delay_s averaged over the 9 episodes of each controller and level

At low demand any adaptive controller cuts stopped delay to under half the fixed plan's; at high demand the fixed plan is within 10% of the best adaptive controller and beats two of them.

Calibration and realism

The twin was tuned to reproduce the observed congestion, not the observed counts: at the chosen scales it carries 22% to 35% of the vehicles counted on each approach, yet its queues reach the published VISSIM queue lengths on the north leg and come within a quarter on the east and south legs. Demand was fitted per approach by running the Dishub fixed plan at a grid of volume scales and taking the scale whose peak-hour maximum queue matched the VISSIM maximum queue from Sumabrata, Haqueline and Mauramdha (2026). The queue is measured the VISSIM way: vehicles below 5 km/h, contiguous from the stop line with gaps up to 20 m.



data/fatmawati/calibration.json · verification over seeds 11 to 14

Only the north approach converged. Near capacity an approach's hourly maximum queue tips from a short queue to spill-back past the 300 m camera and varies strongly between seeds (east: SD 59 m), so the damped fitting rounds overshot and the final scales were bracketed and checked over four seeds instead. The west approach (Jl. RA Kartini) is capacity-limited: above a scale of about 0.38 its demand backs up upstream of the camera, so its 155.5 m target is unreachable and it runs at 0.36 of the observed volume.

Approach	Scale	Twin entry volume (veh/h)	Observed (veh/h)	Share carried	Twin stopped delay (s/veh)	PKJI 2023 control delay (s/veh)
N	0.242	1,219	5,456	22%	191	80
E	0.239	2,269	9,455	24%	99	79
S	0.220	1,008	4,632	22%	86	106
W	0.360	2,219	6,334	35%	42	78

The gap in volume is a SUMO property: its motorcycles discharge far less densely than Jakarta's, so the same queue forms from far fewer vehicles. Read the dataset as a faithful picture of queue dynamics and signal behaviour at this junction, and read its absolute counts, flows and PCU values as a scaled-down traffic stream. Stopped delay is also measured differently from the PKJI control delay (standing time in the camera zone per stop-line vehicle versus the PKJI formula), so the two columns are indicative, not a like-for-like check.

Other calibrated or sourced inputs: the vehicle mix of 80% motorcycles, 19% cars, 0.5% buses and 0.5% trucks follows north-leg counts of 77% to 82% motorcycles (Yusuf and Isradi 2025); the Dishub stage greens of 80, 12, 78 and 45 s come from the same Sumabrata study; PCU factors follow PKJI 2023. Motorcycle dimensions (2.0 x 0.8 m),

acceleration, headway and the 0.5 m standstill and lateral gaps are modelling assumptions, the gaps widened from the VISSIM 0.2 m and 0.3 m because the narrower values produced 15 times more lane collisions at the same queue fit.

Known limitations and caveats

The twin reproduces congestion with ideal sensors, not the real traffic stream, and several columns need more care than the data card gives them. The points marked (recomputed) were found by re-deriving the tables for this report and are not stated in `DATACARD.md`.

Realism of the traffic

- Volumes are 22% to 35% of the field counts per approach (26% overall). The peak-hour maximum queue was the only fitted quantity, and only the north approach met its target.
- Stopped delay differs from the PKJI 2023 control delay by factors from 0.53 (W) to 2.39 (N), and the north value ranges from 100 to 281 s/veh across the four check seeds.
- The `design` level is not the calibrated peak hour (recomputed): the daily profile scales the recorded hours to 0.86 (07:00), 0.64 (12:00) and 0.93 (17:00) of it, so only pm_peak at high demand exceeds the calibrated volume, and the warm-up runs at the lighter preceding hour.
- Windows differ only by that multiplier: the same turning shares, the same vehicle mix and the same 6 ambulances per hour at 07:00, 12:00 and 17:00. There is no tidal change of direction.
- Not modelled: pedestrians, weather, incidents, lane closures, and any driver reaction to a siren beyond one car moving a lane toward the kerb when SUMO allows it. Lanes are a uniform 3.25 m. JORR toll traffic and MRT trains exist only in the 3D viewer.

Sensor idealisation

- Sensors are exact and entry-to-exit re-identification is perfect; a real installation adds detection misses, occlusion and ANPR errors.
- `occupancy_pct` exceeds 100% in 0.67% of loop-minutes, up to 300% (recomputed): it averages SUMO's per-step occupancy, which counts motorcycles riding side by side on one loop.
- Queue readings are censored at the camera's 300 m: `q_m` peaks at 310 m and the north queue is at or beyond 290 m in 4.2% of its ticks (11.4% at high demand), so north maximum queues at and above design demand are not measurable.

- Exit cameras XC_N and XC_S never record a standing vehicle (recomputed), so `exit_q_pcu_N` and `exit_q_pcu_S` are constant zero and the exit-queue term of `pressure_N` and `pressure_S` never acts.
- Exit-gate detections exceed matched passages by 1.7% (554,155 against 544,861), and 96 exits had no entry record although the dictionary says this should be 0.

Labels

- `y_teacher_stage` is a rule, not an optimum: plain `max_pressure` beats `priority_mp` on stopped delay in 12 of 27 paired scenarios and the fixed plan beats it in 7; at pm_peak and high demand every adaptive controller is worse than the fixed plan (138 s/veh against 148 to 200). On ambulance travel time the teacher is slower than `max_pressure` in 8 of 27 scenarios.
- The label is the request, not what the signal did: the 5 s minimum green, 90 s maximum green and the 240 s starvation guard can all override it.
- `controller_request` is not an exact copy of the label under `priority_mp` (recomputed): they agree on 65% of ticks, 95% if the running green stage is also counted, because the request is refreshed only at decision instants while the label is recomputed every tick. It still leaks the label and stays excluded.

Ambulances

- No ambulance ever approaches from the south, so every `ev_*_S` column is zero or empty in all 77,760 ticks and stage S4 is never requested for an ambulance (recomputed).
- The 308 trips detected on E take the free-left slip and never need the signal; only the north and west origins exercise the priority rule.
- West-origin ambulances cross the W stop line, turn around east of the junction, pass the E entry gate and take the slip to the hospital, so for those 206 trips `t_stopline` precedes `t_entry_gate` and `entry_leg` reads E.
- Ambulance records run past the recorded hour: 121 of 732 trips arrive after t = 4,500 s (latest 5,325.5 s). The 8 trips that never arrived, all at pm_peak and high demand, are left out of `ambulance_tt_mean_s`, which flatters those six episodes. The longest trip (2,938 s) spent 2,596 s upstream of the camera zones.
- `recovery_s` is 0 for 31% of trips and empty for 45%; it measures a return to the queue seen at detection, a different definition from the evaluation grid's paired recovery.

Simulation artefacts (`simulation_qc.csv`)

- Over the 108 hours: 373 lane collisions (0.09 per 1,000 veh-km, 115 within 300 m of the junction), 23,976 junction overlaps, 2,019 teleports (2.4 per 1,000 departed vehicles) and 96 unmatched exits. All rise with demand (teleports 129 at low, 727 at design, 1,163 at high) and track stopped delay (Spearman 0.87).

- Nine episodes have junction stopped delay above 150 s/veh, seven of them pm_peak at high demand; the south approach reaches 353 s/veh in episode 98. No episode ends early.
- The smoke test and the `sumo_world.py` docstring still treat lane collisions and unmatched exits as invariants that must be zero; the release does not meet that.

Documentation drift

- `docs/DECISIONS.md` 13.2 still gives the VISSIM motorcycle gaps of 0.2 m and 0.3 m; the shipped vehicle types use 0.5 m and 0.5 m (13.11). 13.1 says the two crossings are 55 m apart; the built network has them 51 m apart.
- The dictionary overstates `controller_request` and `unmatched_exits` (above) and has the typos `buss` and `emergencys`; the data card does not mention the 30 s overrun, the empty south ambulance columns or the role taxonomy.

Licence. The road network and sensor positions derive from OpenStreetMap, so the dataset is shared under the ODbL 1.0 (share-alike, attribution). The simulator is Eclipse SUMO (EPL 2.0); the calibration inputs are CC BY-NC 4.0 (Sumabrata, Haqueline and Mauramdha 2026) and CC BY 4.0 (Yusuf and Isradi 2025). No TomTom or Google data were stored or used.

Intended use

The release serves two jobs in the project: a paired comparison of signal controllers from `episodes.csv`, and supervised learning from `ticks_5s.csv` and `ambulance_trips.csv`. The ambulance evaluation that the methodology scores controllers on is a separate grid produced by the same twin, not part of this release.

Controller comparison. Within each window, level and seed the four controllers saw identical trips, so differences in `episodes.csv` are paired differences, not noise between runs. The outcomes are all computed from the sensor tables: stop-line counts and stopped delay per approach, the junction stopped delay, the fairness ratio (largest over smallest approach delay), the cordon travel time from re-identified gate passages, and the ambulance travel time and queue recovery. The Dishub fixed plan is the today baseline; `max_pressure` is the adaptive reference; `priority_mp` is the teacher the learning stage imitates.

Supervised learning (`traffic_twin/fatmawati/learn.py`, methodology section 13). Features are the `ticks_5s` columns whose role is `feature` in `data_dictionary.csv`; splits keep whole episodes and seeds together.

Task	Label	Source table	Loss	Baseline it must beat
A: imitate the priority-aware teacher	`y_teacher_stage` (S1 to S4)	`ticks_5s`	weighted multinomial log-loss	uniform and frequency-weighted random, the majority stage, and stillness (keep the current stage)
B: forecast each approach queue 30 s ahead	`y_q_veh_30s_N/E/S/W`	`ticks_5s`	Poisson	persistence (the queue now) and the training mean
C: seconds until a detected ambulance crosses the stop line	derived from `t_detect` and `t_stopline`	`ambulance_trips`	quantile loss with tau below 0.5, so the model errs early	the ETA the detector already reports

Before training, validators check the schema and the 5 s spacing, vehicle conservation at the gates, that both labels recompute from the features (the 30 s shift and the teacher rule), and a leakage audit; a shuffled-label run accompanies Task A. A saved Task A model becomes a controller through `load_policy`, so it can be run inside the twin as `learned:<path>` and judged on the same sensors as the hand-written controllers.

Ambulance evaluation and the healthcare framing. Every episode in the release carries ambulance trips to RSUP Fatmawati (732 in all) and the priority-request detector columns, which is what Task A and Task C learn from. The formal scoring in `evaluate.py` and `kpi.py` uses a different protocol: an ambulance released every 10 min at a fixed point at least 400 m before the stop line, cycling through three hospital movements (north through, east free-left, and the west detour via the U-turn) and eight slices of the 231 s cycle, each run paired with the same run without the ambulance and with an empty-road reference. Its outputs (`data/fatmawati_eval*/passages.csv`, `episodes_eval.csv`,

`evaluation_qc.csv`) feed the ambulance KPI, the general-traffic KPI relative to the fixed plan, the safety and cost gates, and the Nash bargaining step. Those files are evaluation outputs, not sensor data, and are documented in `docs/DECISIONS.md` section 13.10.

Quick start

The tables load with any CSV reader; join them on `episode_id`, and pick feature columns from `data_dictionary.csv` rather than by hand so the excluded columns stay out.

Python

```
import pandas as pd

d = "data/fatmawati_v3"
eps = pd.read_csv(f"{d}/episodes.csv")
ticks = pd.read_csv(f"{d}/ticks_5s.csv", low_memory=False)
ticks = ticks.merge(eps[["episode_id", "window", "level", "seed"]],
on="episode_id")

dd = pd.read_csv(f"{d}/data_dictionary.csv")
feats = dd.query("table == 'ticks_5s' and role == 'feature'")
["column"].tolist()
X, y = ticks[feats], ticks["y_teacher_stage"]

cams = pd.read_csv(f"{d}/cameras_1min.csv").merge(pd.read_csv(f"
{d}/sensors.csv"), on=["sensor_id", "kind", "leg"])
```

R

```
library(readr); library(dplyr)

d <- "data/fatmawati_v3"
eps <- read_csv(file.path(d, "episodes.csv"))
ticks <- read_csv(file.path(d, "ticks_5s.csv"), guess_max = 100000) |>
  left_join(select(eps, episode_id, window, level, seed), by = "episode_id")
dd <- read_csv(file.path(d, "data_dictionary.csv"))
feats <- dd |> filter(table == "ticks_5s", role == "feature") |> pull(column)
```

Things that trip readers up:

- Empty cells are missing values, not zeros: `speed_kmh` when no vehicle crossed the loop, `cam_speed_kmh_*` when a zone is empty, `ev_dist_m_*` and the other ambulance fields when no ambulance is in the zone, `y_q_veh_30s_*` never (every tick has a forecast label), `t_arrive` for ambulances still travelling.

- `t` and `t_start` include the 15 min warm-up, so recorded time runs from 900 to 4,500 s; `clock` is a text column.
- `ticks_5s.csv` already has `controller`; do not merge it again from `episodes.csv`.
- `controller_request` is the teacher's held request under `priority_mp`; keep it, `controller`, `episode_id`, `t`, `clock` and every `y_*` column out of the features.
- `arrived` in `ambulance_trips.csv` is written as `True` / `False`.
- Group splits by `episode_id` or `seed`, never by row: rows 5 s apart in the same episode are nearly identical.

Reproduce the release. Needs SUMO 1.27.1 with libsumo and sumolib (`deploy/gcp/requirements-twin.txt`). The release takes about 2 h on 8 workers; `--plan smoke` runs one 3-minute episode to check the installation.

```
python3 -m traffic_twin.fatmawati.build_net
```

```
python3 -m traffic_twin.fatmawati.calibrate --workers 8 --start
'{"N":0.242,"E":0.239,"S":0.22,"W":0.36}'
```

```
python3 -m traffic_twin.fatmawati.generate --plan release --out
data/fatmawati_v3 --workers 8
```

```
shasum -a 256 -c data/fatmawati_v3/SHA256SUMS.txt
```

```
python3 -m pytest -q tests/test_fatmawati.py
```